

Information and Communication Technology

Everyday life for employees, students, consumers, savers, investors, managers, and entrepreneurs is changing and becoming more complex all the time. Although rapid changes in information and communication technology have an impact on all aspects of life, the operations of business are particularly affected. The information and communication technology courses aim to prepare students for a business and communication world that relies more and more on electronic technology, technology that changes at a rapid pace. Whether students wish to apply the new technologies in their careers in the workplace, to develop a broad understanding of the effects of technology on business, or to establish a foundation for the use of information technology at the postsecondary level, the information and communication technology courses will provide opportunities to develop relevant skills and knowledge.

Strands

Information and Communication Technology: The Digital Environment, Grade 11, Open

- Digital Literacy
- Productivity Software
- Business Communications
- E-Business
- Information and Communication Technology Ethics and Issues

Information and Communication Technology: Multimedia Solutions, Grade 12, College Preparation

- The Electronic Business Environment
- Applications Software
- Electronic Communications
- Project Management

Information and Communication Technology in the Workplace, Grade 12, Workplace Preparation

- The Electronic Workplace Environment
- Applications Software
- Electronic Communications and Teamwork
- Preparation for Employment

Information and Communication Technology (BTX4E) in the Workplace, Grade 12, Workplace Preparation

This course provides students with the opportunity to further develop essential workplace skills in information and communication technology while working in a team environment. Using a project-based approach, students will focus on integrating software applications and applying multimedia software features. Students will expand their understanding of electronic business and e-commerce environments and workplace ethics. This course will prepare students for a successful transition from secondary school to the workplace.

Prerequisite: Information and Communication Technology: The Digital Environment, Grade 11, Open

The Electronic Workplace Environment

Overall Expectations

By the end of this course, students will:

- describe workplace settings from an information and communication technology perspective;
- explain the impact of information and communication technology on the workplace environment;
- explain the importance of adhering to ethical standards when conducting business electronically.

Specific Expectations

The Digital Workplace

By the end of this course, students will:

- identify a variety of workplace settings (e.g., home office, small business, large corporation);
- explain the ways in which workplace settings have changed as a result of recent developments in information and communication technology;
- identify types of computer networks;
- describe the hardware and software required for a specific workplace setting (e.g., health care facility, bank, travel agency).

The Impact of Information and Communication Technology

By the end of this course, students will:

- list the positive and negative effects of information and communication technology on working conditions (e.g., telecommuting, job reassignment, flex-time, employee monitoring);

- explain how information and communication technology affects the way in which business is conducted (e.g., more efficient communication, ease of information sharing and retrieval, reduced mailing costs);
- identify different types of businesses (e.g., e-commerce, data warehouse, web host) that have evolved as a result of information and communication technology developments.

Workplace Ethics

By the end of this course, students will:

- explain why it is important to have a personal code of ethics;
- identify important elements of a business code of ethics;
- describe the consequences of unethical behaviour related to information and communication technology for an employee and for a business.

Applications Software

Overall Expectations

By the end of this course, students will:

- use appropriate software to create integrated customized documents that meet professional business standards;
- demonstrate the effective use of multimedia software;
- use appropriate software to facilitate the completion of a team project that involves a multimedia product.

Specific Expectations

Integrating Applications

By the end of this course, students will:

- identify the most appropriate software application for specific tasks;
- select the most appropriate software application (e.g., database, graphic editing, animation, concept mapping) for creating customized business documents that meet professional business standards;
- demonstrate appropriate use of advanced software features and functions (e.g., styles, autocorrect/quick correct, templates, object linking, payment function, database reports) to create business documents that meet professional business standards;
- produce integrated business documents (e.g., reports with charts, mail merge letters using data in an existing data source, linked documents in a presentation) for specific purposes.

Multimedia

By the end of this course, students will:

- differentiate between multimedia file formats (e.g., GIF, JPG, PNG, WAV, AVI) and identify the applications with which each format is compatible;

- use graphic editing software to edit (e.g., crop; adjust colour, size) and export graphics;
- use multimedia features (e.g., sound, animation, video, graphics) to enhance a product (e.g., web page, presentation, e-newsletter).

Using Software for a Team Project

By the end of this course, students will:

- use appropriate software to organize the tasks involved in completing a team project (e.g., setting goals, organizing tasks, scheduling, budgeting, controlling, evaluating);
- demonstrate the use of electronic tools (e.g., file sharing, e-mail) to facilitate the completion of the team project;
- create a multimedia product in a team environment, using appropriate software;
- communicate the finished product to an appropriate audience (e.g., present and/or demonstrate the product; provide the opportunity for the audience to try out the product and to provide feedback on its effectiveness).

Electronic Communications and Teamwork

Overall Expectations

By the end of this course, students will:

- use a variety of electronic resources to retrieve, evaluate, and communicate information;
- demonstrate an understanding of the team process in the work environment;
- use software and electronic tools to implement organizational and time-management strategies.

Specific Expectations

Electronic Research and Communication

By the end of this course, students will:

- use a variety of search engines to locate information for specific purposes;
- evaluate information collected from electronic sources for usefulness, validity, bias, accuracy, reliability, and currency;
- adhere to copyright and licensing rules and regulations;
- identify ways of communicating electronically with individuals working in different locations;
- use the appropriate type of electronic tool for communicating a message to a specific audience.

The Team Process

By the end of this course, students will:

- explain why businesses form teams to complete information and communication technology projects;

- identify personal qualities that enhance employee effectiveness and productivity in a team environment (e.g., promptness, ability to get along with others, dependability, honesty, willingness to ask questions, respect for diversity, positive attitude);
- identify ways of working effectively in a team with co-workers who are of different ages, come from different cultures, and have a range of abilities.

Organization and Time Management

By the end of this course, students will:

- organize computer files into appropriately named folders;
- explain ways in which software and electronic tools can help with organization and time management;
- demonstrate the ability to use software for electronic time management (e.g., create and maintain a to-do list, a calendar, a planner).

Preparation for Employment

Overall Expectations

By the end of this course, students will:

- demonstrate an understanding of the importance of keeping an updated digital portfolio that accurately reflects their information and communication technology competencies and skills for job interview purposes;
- demonstrate an understanding of strategies for conducting an effective job search;
- develop strategies to make an effective transition from school to career.

Specific Expectations

Digital Portfolio

By the end of this course, students will:

- identify employability skills and essential skills by accessing appropriate websites (e.g., Conference Board of Canada, Human Resources and Social Development Canada) and prepare a list of these skills for inclusion in their portfolio;
- classify their information and communication technology skills (e.g., skills in web-page design, creating multimedia presentations, desktop publishing) and include a summary of them in their portfolio;
- revise their digital portfolio to include samples of their current exemplary work.

The Job Search

By the end of this course, students will:

- identify employment opportunities that require information and communication technology skills;
- identify the stages of the interview process;

- update documents for the purposes of a job search (e.g., résumé, letter of application, follow-up letter);
- describe effective job search techniques and strategies (e.g., scanning want ads, applying online, conducting a targeted mailing, attending career fairs, networking).

School-to-Career Transition

By the end of this course, students will:

- identify resources (e.g., books, conferences, continuing education courses, distance learning, employer-provided training) that can be used to develop and strengthen workplace skills;
- identify financial planning resources (e.g., currency exchange rate calculator, car payment calculator, savings growth calculator) available to assist with financial planning;
- prepare a financial plan to support the transition from school to career.